

COURSE: CS 356 — AI Ethics and Fairness

Credits: 3 | Department: Computer Science

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PREREQUISITES: CS 351

CO-REQUISITES: CS 352 (may be taken concurrently — co-requisite, not prerequisite)

OFFERED: Spring

DESCRIPTION:

Ethical dimensions of artificial intelligence systems. Topics include algorithmic bias and fairness (demographic parity, equalized odds, calibration), privacy (differential privacy, federated learning), accountability, transparency and explainability (LIME, SHAP), AI governance and regulation, environmental costs of ML, and case studies (facial recognition, credit scoring, hiring algorithms). Reading-heavy course with significant written work.

GRADING: Weekly reading responses 20%, Case study analyses 30%, Final paper 30%, Participation 20%.

NOTE: CS 356 can be taken concurrently with CS 352 (Deep Learning). It does NOT require CS 352 to be completed first, but CS 351 (Machine Learning Foundations) must be completed before enrollment.